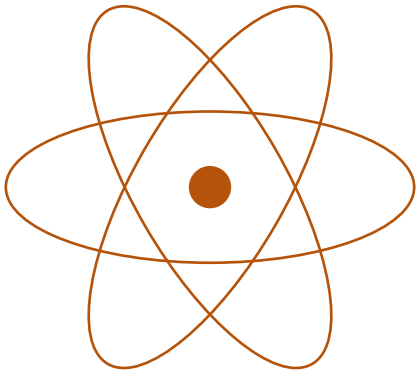
SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Units Conversion & Dimensional Analysis

Detailed Solutions Manual

Olympiad Training • Physics Fundamentals

SPhOH Team

FIRST EDITION • 2026

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Preface

This document provides complete, step-by-step mathematical and physical solutions for all problems presented in the **Units Conversion and Dimensional Analysis** problem set (`UCA-I`) published by the **Saudi Physics Olympiad Hub (SPhOH)**.

Rigorous dimensional derivations, unit conversions, Buckingham Π analysis, and conceptual justifications are detailed for each question.

CHAPTER 1

Units Conversion Solutions

1.1 Multiple Choice Questions

SOLUTION 1 LEVEL I

Temperature Difference Conversion

Correct Option: (b) 18°F
Detailed Explanation:
The absolute temperature transformation between Celsius (C) and Fahrenheit (F) is $F = \frac{9}{5}C + 32$. However, when considering a *temperature difference* ΔT, the constant offset +32°F cancels out completely:

$$\Delta T_{\mathrm{F}} = T_{F,2} - T_{F,1} = \left(\frac{9}{5}T_{C,2} + 32\right) - \left(\frac{9}{5}T_{C,1} + 32\right) = \frac{9}{5}(T_{C,2} - T_{C,1}) = \frac{9}{5}\Delta T_{\mathrm{C}}$$

Substituting $\Delta T_{\mathrm{C}} = 10°\mathrm{C}$:

$$\Delta T_{\mathrm{F}} = \frac{9}{5} \times 10 = 18°\mathrm{F}$$

SOLUTION 2 LEVEL I

Engine Rotational Speed to Angular Velocity

Correct Option: (b) 20 **rad/s**
Detailed Explanation:
To convert rotational speed from revolutions per minute (RPM) to radians per second (rad/s), we multiply by the unit conversion factors: 1 revolution = 2π radians and 1 minute = 60 seconds.

$$\omega = \left(\frac{600}{\pi}\frac{\text{rev}}{\text{min}}\right) \times \left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right) \times \left(\frac{1 \text{ min}}{60 \text{ s}}\right) = \frac{600 \times 2\pi}{\pi \times 60} = \frac{1200}{60} = 20 \text{ rad/s}$$

SOLUTION 3 LEVEL II

Density Conversion to Base SI Units

Correct Option: (c) $1,000$ **kg/m**^{3}
Detailed Explanation:
We convert the numerator (mg $\rightarrow$ kg) and denominator (mm^3 $\rightarrow$ m^3) separately:

$$1\ \text{mg} = 10^{-6}\ \text{kg}$$
$$1\ \text{mm}^3 = (10^{-3}\ \text{m})^3 = 10^{-9}\ \text{m}^3$$

Substituting these into the given density expression:

$$1\ \text{mg/mm}^3 = \frac{10^{-6}\ \text{kg}}{10^{-9}\ \text{m}^3} = 10^{-6-(-9)}\ \text{kg/m}^3 = 10^3\ \text{kg/m}^3 = 1,000\ \text{kg/m}^3$$

SOLUTION 4 LEVEL II

Fuel Efficiency Unit Conversion

Correct Option: (c) 5 **L**/100**km**
Detailed Explanation:
The fuel efficiency is given as 20 km/L. Inverting this quantity yields fuel consumption per single kilometer:

$$\frac{1}{20\ \text{km/L}} = 0.05\ \text{L/km}$$

To express fuel consumption in liters per 100 km, we multiply the distance in the denominator by 100:

$$0.05\ \text{L/km} = 0.05 \times 100\ \text{L/100km} = 5\ \text{L/100km}$$

SOLUTION 5 LEVEL II

Conversion Factor for Acceleration

Correct Option: (c) Divide by 1000, then multiply by 3600
Detailed Explanation:
We convert acceleration from $\mathrm{m/s^2}$ to $\mathrm{km/min^2}$:

$$1\ \mathrm{m/s^2} = \left(\frac{1\ \mathrm{m}}{1\ \mathrm{s^2}}\right) \times \left(\frac{1\ \mathrm{km}}{1000\ \mathrm{m}}\right) \times \left(\frac{60\ \mathrm{s}}{1\ \mathrm{min}}\right)^2 = \frac{1}{1000} \times 3600\ \mathrm{km/min^2}$$

Thus, to perform the conversion, one must divide by 1000 (to convert meters to kilometers) and multiply by $3600 = 60^2$ (to convert inverse squared seconds to inverse squared minutes).

SOLUTION 6

LEVEL III

Kinematic Viscosity in CGS System

Correct Option: (a) 1.5×10^{-2} **St**
Detailed Explanation:
First, convert dynamic viscosity μ into CGS base units ($\mathrm{g \cdot cm^{-1} \cdot s^{-1}}$):

$$\mu = 1.2\ \mathrm{cP} = 1.2 \times 10^{-3}\ \mathrm{Pa \cdot s} = 1.2 \times 10^{-3}\ \mathrm{kg \cdot m^{-1} \cdot s^{-1}}$$

Converting kg/m to g/cm:

$$1.2 \times 10^{-3} \frac{\mathrm{kg}}{\mathrm{m \cdot s}} \times \frac{1000\ \mathrm{g}}{1\ \mathrm{kg}} \times \frac{1\ \mathrm{m}}{100\ \mathrm{cm}} = 1.2 \times 10^{-2}\ \mathrm{g \cdot cm^{-1} \cdot s^{-1}}$$

The density is already given in CGS units: $\rho = 0.80\ \mathrm{g/cm^3}$. Now compute the kinematic viscosity $\nu = \frac{\mu}{\rho}$:

$$\nu = \frac{1.2 \times 10^{-2}\ \mathrm{g \cdot cm^{-1} \cdot s^{-1}}}{0.80\ \mathrm{g \cdot cm^{-3}}} = 1.5 \times 10^{-2}\ \mathrm{cm^2/s} = 1.5 \times 10^{-2}\ \mathrm{St}$$

1.2 Free-Response Questions

SOLUTION 7

LEVEL I

Intravenous Fluid Flow Rate

Given flow rate $R = 2$ fl oz/hr and 1 fl oz ≈ 29.57 cm^3.
Using the chain-link conversion method:

$$R = 2\frac{\text{fl oz}}{\text{hr}} \times \left(\frac{29.57\ \text{cm}^3}{1\ \text{fl oz}}\right) \times \left(\frac{1000\ \text{mm}^3}{1\ \text{cm}^3}\right) \times \left(\frac{1\ \text{hr}}{3600\ \text{s}}\right)$$

$$R = \frac{2 \times 29.57 \times 1000}{3600}\ \text{mm}^3/\text{s} = \frac{59140}{3600}\ \text{mm}^3/\text{s} \approx 16.43\ \text{mm}^3/\text{s}$$

SOLUTION 8

LEVEL II

Alien Unit Conversion for Force

Alien system base unit equivalents:

- $1\ \text{Z} = 1.5\ \text{kg} \implies 1\ \text{kg} = \frac{1}{1.5}\ \text{Z} = \frac{2}{3}\ \text{Z}$
- $1\ \text{B} = 0.5\ \text{m} \implies 1\ \text{m} = \frac{1}{0.5}\ \text{B} = 2\ \text{B}$
- $1\ \text{G} = 2\ \text{s} \implies 1\ \text{s} = \frac{1}{2}\ \text{G} = 0.5\ \text{G}$

Force in SI base units is $F = 10\ \text{N} = 10\ \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$.
Substituting the unit conversion factors:

$$F = 10 \times \left(\frac{2}{3}\ \text{Z}\right) \times (2\ \text{B}) \times (0.5\ \text{G})^{-2}$$

$$F = 10 \times \frac{4}{3}\ \text{Z} \cdot \text{B} \times \frac{1}{0.25\ \text{G}^2} = 10 \times \frac{4}{3} \times 4 = \frac{160}{3}\ \text{Z} \cdot \text{B/G}^2 \approx 53.33\ \text{Z} \cdot \text{B/G}^2$$

SOLUTION 9

LEVEL II

van der Waals Constant Conversion

Given $a = 0.138\ \text{Pa} \cdot \text{m}^6/\text{mol}^2$. Conversion relations:

$$1\ \text{Pa} = \frac{1}{101325}\ \text{atm}, \quad 1\ \text{m}^3 = 10^3\ \text{L} \implies 1\ \text{m}^6 = (10^3\ \text{L})^2 = 10^6\ \text{L}^2$$

Substituting into a:

$$a = 0.138 \times \left(\frac{1}{101325}\ \text{atm}\right) \times (10^6\ \text{L}^2) \times \text{mol}^{-2}$$

$$a = \frac{138000}{101325}\ \text{atm} \cdot \text{L}^2/\text{mol}^2 \approx 1.362\ \text{atm} \cdot \text{L}^2/\text{mol}^2$$

SOLUTION 10

LEVEL II

Volume of Observable Universe in Cubic Parsecs

First, convert radius $r = 46.5 \times 10^9$ ly to parsecs using 1 pc = 3.26 ly:

$$r = \frac{46.5 \times 10^9\ \text{ly}}{3.26\ \text{ly/pc}} \approx 1.4264 \times 10^{10}\ \text{pc}$$

Calculating spherical volume $V = \frac{4}{3}\pi r^3$:

$$V = \frac{4}{3}\pi(1.4264 \times 10^{10}\ \text{pc})^3 \approx 4.1888 \times (2.902 \times 10^{30}\ \text{pc}^3) \approx 1.215 \times 10^{31}\ \text{pc}^3$$

SOLUTION 11

LEVEL II

Stefan-Boltzmann Constant in CGS System

Given $\sigma = 5.67 \times 10^{-8}\ \text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$.
1. **SI Fundamental Units:** Since 1 W = 1 J/s = 1 kg · m^2 · s^{-3}:

$$[\sigma] = \frac{\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-3}}{\text{m}^2 \cdot \text{K}^4} = \text{kg} \cdot \text{s}^{-3} \cdot \text{K}^{-4}$$

Notice that length cancels out completely from the fundamental unit expression.
2. **Conversion to CGS System:** Converting 1 kg = 1000 g:

$$\sigma_{\text{CGS}} = 5.67 \times 10^{-8} \times \left(\frac{1000\ \text{g}}{1\ \text{kg}}\right) = 5.67 \times 10^{-5}\ \text{g} \cdot \text{s}^{-3} \cdot \text{K}^{-4}$$

SOLUTION 12

LEVEL III

Planck Length in Natural Units

In natural units ($\hbar = c = G = 1$), energy E is measured in eV or GeV. We convert meter to GeV^{-1}.
First, calculate the product $\hbar c$:

$$\hbar c = (1.05 \times 10^{-34}\ \text{J} \cdot \text{s}) \times (3.0 \times 10^{8}\ \text{m/s}) = 3.15 \times 10^{-26}\ \text{J} \cdot \text{m}$$

Convert Joules to GeV ($1\ \text{GeV} = 10^{9} \times 1.6 \times 10^{-19}\ \text{J} = 1.6 \times 10^{-10}\ \text{J}$):

$$\hbar c = \frac{3.15 \times 10^{-26}}{1.6 \times 10^{-10}}\ \text{GeV} \cdot \text{m} \approx 1.96875 \times 10^{-16}\ \text{GeV} \cdot \text{m}$$

Since $\hbar c = 1$ in natural units, we obtain the unit equivalence:

$$1\ \text{m} = \frac{1}{1.96875 \times 10^{-16}}\ \text{GeV}^{-1} \approx 5.08 \times 10^{15}\ \text{GeV}^{-1}$$

Now convert Planck length $l_P \approx 1.6 \times 10^{-35}$ m:

$$l_P = (1.6 \times 10^{-35}\ \text{m}) \times (5.08 \times 10^{15}\ \text{GeV}^{-1}/\text{m}) \approx 8.13 \times 10^{-20}\ \text{GeV}^{-1}$$

CHAPTER 2

Dimensional Analysis Solutions

2.1 Multiple Choice Questions

SOLUTION 13 **LEVEL I**

Condition for Addition of Physical Quantities

Correct Option: (b) The dimensions of X, Y, and Z must be exactly identical.
Detailed Explanation:
According to the Principle of Dimensional Homogeneity, physical quantities can only be added or subtracted if they share the exact same physical dimensions. Thus, in $X = Y + Z$, we must have $[X] = [Y] = [Z]$.

SOLUTION 14 **LEVEL I**

Dimensional Verification of Time Equations

Correct Option: (c) $t = d \cdot v$
Detailed Explanation:
Checking dimensions for each option ($[d] = L$, $[v] = LT^{-1}$, $[t] = T$):

- (a) $[d/v] = L/(LT^{-1}) = T$ (Valid dimension of time)
- (b) $[2d/v] = L/(LT^{-1}) = T$ (Valid dimension of time)
- (c) $[d \cdot v] = L \times (LT^{-1}) = L^2T^{-1} \neq T$ (Dimensionally impossible)

SOLUTION 15 **LEVEL I**

Dimension of Wavenumber k

Correct Option: (b) L^{-1} (Inverse of length)
Detailed Explanation:
The argument of any trigonometric function (like sin) must be dimensionless: $[k \cdot x] = 1$. Since $[x] = L$, we have:

$$[k] \cdot L = 1 \implies [k] = L^{-1}$$

SOLUTION 16

LEVEL I

Dimension of Pure Constant π

Correct Option: (c) Dimensionless (1)
Detailed Explanation:
Pure numerical constants (π, e, $\sqrt{2}$, fractions) have no physical dimensions: $[\pi] = 1$.

SOLUTION 17

LEVEL I

Unit of Spring Constant k

Correct Option: (b) Newton / meter
Detailed Explanation:
From Hooke's Law $F = k \cdot x$, isolating k gives $k = \frac{F}{x}$. Thus, the SI unit is N/m.

SOLUTION 18

LEVEL I

Dimensions of Volumetric Flow Rate

Correct Option: (c) L^3T^{-1}
Detailed Explanation:
Drainage rate R is volume per unit time $R = \frac{\Delta V}{\Delta t}$. Dimensions are:

$$[R] = \frac{[V]}{[t]} = \frac{L^3}{T} = L^3T^{-1}$$

SOLUTION 19 LEVEL I

Relationship Between Work and Kinetic Energy Dimensions

Correct Option: (c) They have exactly the same dimensions.
Detailed Explanation:
Work: $[W] = [F][d] = (MLT^{-2})(L) = ML^2T^{-2}$.
Kinetic Energy: $[K] = [m][v]^2 = M(LT^{-1})^2 = ML^2T^{-2}$.
Both quantities represent energy and share identical dimensions.

SOLUTION 20 LEVEL I

Identification of Dimensionless Quantity

Correct Option: (a) Angle θ
Detailed Explanation:
Angle in radians is arc length divided by radius: $[\theta] = \frac{L}{L} = 1$. Acceleration has dimensions LT^{-2} and density has dimensions ML^{-3}.

SOLUTION 21 LEVEL I

Physical Quantity with Dimensions LT^{-2}

Correct Option: (b) Acceleration
Detailed Explanation:
Velocity is LT^{-1}, acceleration is LT^{-2}, and force is MLT^{-2}.

SOLUTION 22 LEVEL II

Dimension of Exponential Constant c

Correct Option: (b) T^{-1} (Inverse of time)
Detailed Explanation:
Exponents in exponential functions e^{-ct} must be dimensionless: $[c \cdot t] = 1$.

$$[c] \cdot T = 1 \implies [c] = T^{-1}$$

SOLUTION 23

LEVEL III

Chandrasekhar Mass Limit Proportionality

Correct Option: (a) $M_{\text{Ch}} \propto \left(\frac{\hbar c}{G}\right)^{3/2} m_n^{-2}$
Detailed Explanation:
Set $M_{\text{Ch}} \propto \hbar^a c^b G^c m_n^d$. Dimensions of constants:

$$[\hbar] = ML^2T^{-1}, \quad [c] = LT^{-1}, \quad [G] = M^{-1}L^3T^{-2}, \quad [m_n] = M$$

Dimensional equation for mass $[M^1L^0T^0]$:

$$M^1L^0T^0 = (ML^2T^{-1})^a (LT^{-1})^b (M^{-1}L^3T^{-2})^c M^d$$

Equating exponents for M, L, T:

$$M: \quad a - c + d = 1 \tag{2.1}$$
$$L: \quad 2a + b + 3c = 0 \tag{2.2}$$
$$T: \quad -a - b - 2c = 0 \tag{2.3}$$

Adding (2.2) and (2.3) yields:

$$a + c = 0 \implies c = -a$$

Substituting $c = -a$ into (2.2):

$$2a + b + 3(-a) = 0 \implies b = a$$

Substituting $c = -a$ into (2.1):

$$a - (-a) + d = 1 \implies 2a + d = 1$$

For electron degeneracy in relativistic white dwarfs, $a = 3/2$, which gives:

$$a = 3/2, \quad b = 3/2, \quad c = -3/2, \quad d = -2$$

Thus:

$$M_{\text{Ch}} \propto \hbar^{3/2} c^{3/2} G^{-3/2} m_n^{-2} = \left(\frac{\hbar c}{G}\right)^{3/2} m_n^{-2}$$

2.2 Free-Response Questions

SOLUTION 24 LEVEL I

Dimensional Inconsistency in Proposed Velocity Equation

Dimensional Analysis:
Let us check the dimensions of each term in $v = d + a \cdot t$:

- Left-hand side: $[v] = LT^{-1}$
- First term on RHS: $[d] = L$
- Second term on RHS: $[a \cdot t] = (LT^{-2})(T) = LT^{-1}$

The equation attempts to add distance (L) to velocity (LT^{-1}).
Physical Conclusion:
This equation is fundamentally impossible because it violates the **Principle of Dimensional Homogeneity**, which strictly states that physical quantities added or subtracted must share identical dimensions.

SOLUTION 25 LEVEL II

Limitations of Dimensional Analysis Regarding Numerical Constants

Scientific Explanation:
Dimensional analysis operates strictly on the fundamental physical dimensions ($M, L, T, I, \ldots$) of physical variables. Pure numerical factors (such as $\frac{1}{2}, 2\pi, \mathrm{e}$) are dimensionless quantities ($[C] = 1$).
Because dimensionless numbers carry zero dimensional units, multiplying or dividing any physical expression by a dimensionless constant leaves the overall dimensional balance completely unchanged. Therefore, dimensional analysis cannot determine numerical prefactors, which must be derived using calculus, physical conservation laws, or empirical measurement.

SOLUTION 26 LEVEL II

Validity of Dimensionally Homogeneous Equations

Logical Argument:
No, dimensional correctness is a **necessary but not sufficient** condition for an equation to be physically valid.
Examples:
1. **Missing dimensionless constants:** The equation $K = mv^2$ is dimensionally correct (ML^2T^{-2}), but physically incorrect because it misses the factor of $\frac{1}{2}$.
2. **Irrelevant terms:** The equation $x = vt + d_0 + \frac{F}{m}$ compares position (L) with force per mass (LT^{-2}), but even if we modified it to $x = vt + \frac{Ft^2}{m}$, it might represent an incorrect force law for a specific setup despite being dimensionally homogeneous.

SOLUTION 27

LEVEL II

Transcendental Functions and Dimensionless Arguments

Physical Justification:
Transcendental mathematical functions ($\sin x, \cos x, \ln x, e^x$) can be expressed as Taylor/Maclaurin power series expansions:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

If x possessed a physical dimension (such as length L), the expansion would sum terms with different dimensions ($L - L^3 + L^5 - \dots$), explicitly violating the Principle of Dimensional Homogeneity.
Physical Impossibility:
Evaluating $\sin(5 \text{ meters})$ or $\log(10 \text{ kg})$ is physically meaningless because trigonometric and logarithmic operations represent pure numerical ratios.

SOLUTION 28

LEVEL III

Ocean Surface Wave Velocity Derivation

(a) Buckingham Π Theorem Analysis:
Variables: $v, \lambda, \rho, g, \gamma$ ($N = 5$). Core dimensions: M, L, T ($K = 3$).
Number of dimensionless Π groups: $N - K = 5 - 3 = 2$.

Selecting repeating variables ρ, g, λ:

$$\Pi_1 = \frac{v}{\sqrt{g\lambda}}, \quad \Pi_2 = \frac{\gamma}{\rho g \lambda^2}$$

The wave dynamics combine two distinct restoring forces: gravity (g) and surface tension (γ). Summing their contributions to wave kinetic energy yields:

$$v^2 = C_1 g\lambda + C_2 \frac{\gamma}{\rho\lambda}$$

Full hydrodynamic derivation yields $C_1 = \frac{1}{2\pi}$ and $C_2 = \frac{2\pi}{1}$:

$$v = \sqrt{\frac{g\lambda}{2\pi} + \frac{2\pi\gamma}{\rho\lambda}}$$

(b) Dominant Regimes:

- **Gravity Waves ($\lambda \to \infty$):** The surface tension term becomes negligible ($\frac{2\pi\gamma}{\rho\lambda} \to 0$), giving $v \approx \sqrt{\frac{g\lambda}{2\pi}}$.
- **Capillary Waves ($\lambda \to 0$):** The surface tension term dominates over gravity, giving $v \approx \sqrt{\frac{2\pi\gamma}{\rho\lambda}}$.

SOLUTION 29

LEVEL III

Nuclear Explosion Blast Wave Radius

(a) Derivation of $R(t)$ via Dimensional Analysis:
Assume $R(t) = CE^a\rho_0^b t^c$. Dimensions:

$$[R] = L, \quad [E] = ML^2T^{-2}, \quad [\rho_0] = ML^{-3}, \quad [t] = T$$

Dimensional balance $[M^0L^1T^0]$:

$$M^0L^1T^0 = M^{a+b}L^{2a-3b}T^{-2a+c}$$

Equating exponents:

$$\begin{aligned} M: &\quad a + b = 0 \implies b = -a \\ L: &\quad 2a - 3b = 1 \implies 2a - 3(-a) = 1 \implies 5a = 1 \implies a = 1/5,\ b = -1/5 \\ T: &\quad -2a + c = 0 \implies c = 2a = 2/5 \end{aligned}$$

Thus, Taylor's blast wave relationship is:

$$R(t) = C\left(\frac{E}{\rho_0}\right)^{1/5} t^{2/5}$$

(b) Radius Calculation:
Since $R(t) \propto t^{2/5}$:

$$\frac{R_2}{R_1} = \left(\frac{t_2}{t_1}\right)^{2/5} = \left(\frac{16.0\ \text{ms}}{1.0\ \text{ms}}\right)^{2/5} = (16)^{2/5} = (2^4)^{2/5} = 2^{8/5} \approx 3.0314$$

Substituting $R_1 = 80$ m:

$$R_2 = 80 \times 3.0314 \approx 242.5\ \text{m}$$